

Introduction to Quantum Computing

First Lab Exercise: QUIRK

March 2025

1 1-qubit quantum gates

1. What are the following quantum gate products equivalent to?
 - (a) $X \cdot X$
 - (b) $H \cdot H$
 - (c) $X^{1/2} \cdot X^{1/2}$
2. Remember that a *global* phase change does not alter a qubit. Build a quantum circuit that, from a single qubit initialized to $|0\rangle$, generates the following 1-qubit states:
 - (a) $|1\rangle$
 - (b) $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$
 - (c) $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$
 - (d) $|R\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle)$

2 Multi-qubit quantum gates

2.1 Control gates: CNOT and others

1. Hover the mouse pointer over the two default qubit lines and check that both qubits are on the $|0\rangle$ state.
2. Drop a NOT gate over the second line and annotate the change on the corresponding Bloch sphere.
3. Drop a **control** gate (small black circle) over the first line and on the same horizontal position that the first Pauli-X (NOT) gate. A new vertical segment joining the **control** and the NOT gate should appear. Annotate any change on any of the Bloch spheres.
4. Change the state of the first qubit to $|1\rangle$ by clicking on the $|0\rangle$ symbol on the left of the corresponding qubit line. Annotate any changes on the Bloch spheres.
5. Repeat the experiment using an **anti-control** (small white circle) instead of the **control**.
6. Repeat the experiment using a Hadamard gate instead of the NOT one.

2.2 Building entangled Bell states

Using a Hadamard gate and a CNOT¹ gate whose control is applied to the output of the Hadamard gate, build a quantum circuit that turns a 2-qubit initial state $|0\rangle \otimes |0\rangle$ to the following Bell states:

1. $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|0\rangle \otimes |0\rangle + |1\rangle \otimes |1\rangle)$
2. $|\Phi^-\rangle = \frac{1}{\sqrt{2}}(|0\rangle \otimes |0\rangle - |1\rangle \otimes |1\rangle)$
3. $|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|0\rangle \otimes |1\rangle - |1\rangle \otimes |0\rangle)$

¹NOT gate with control.

2.3 No qubit cloning, even by the CNOT gate

Because of the No-Cloning Theorem, a process whose effect is copying the whole quantum state of a system into another identical system is not physically possible. Now, let $\mathbb{C}|xy\rangle$ represent a CNOT gate whose first qubit x is the control qubit. Since

$$\mathbb{C}|00\rangle = |00\rangle \quad \mathbb{C}|10\rangle = |11\rangle$$

and, because of linearity,

$$\mathbb{C}(\alpha|00\rangle + \beta|10\rangle) = \alpha|00\rangle + \beta|11\rangle,$$

the CNOT gate might appear to allow the cloning of the first qubit onto the second originally set to 0. This would be a violation of the no-cloning theorem. In order to prove that the CNOT gate does NOT provide a way of violating the no-cloning theorem, answer the following exercises:

1. The 1-qubit state $|+\rangle$ is defined as above in 2b by

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle).$$

Using the base $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$ to describe a 2-qubit quantum state, answer the following questions:

- (a) What is the 2-qubit state if each qubit has been initialized to $|+\rangle$?
 - (b) The first qubit is initialized to $|+\rangle$ and the second, to $|0\rangle$. What is the 2-qubit state if a CNOT gate has been applied targeting the second qubit, the first one being the control?
 - (c) Can you use a CNOT gate for breaking the no-clone theorem? Please, justify your answer on the previous questions of this exercise.
2. Using Hadamard gates, prepare 2 quantum registers, each one composed of 2 qubits. The states of this registers will be:
 - Register A: $|+\rangle \otimes |+\rangle = (H|0\rangle) \otimes (H|0\rangle)$
 - Register B: the first qubit is initialized to $|+\rangle = H|0\rangle$. The second one is initialized by applying a CNOT gate to the quantum state $|0\rangle$. The control of this CNOT gate is the first qubit, previously initialized to $|+\rangle$.

Answer the following questions:

- (a) Compute the probability of measuring a $|1\rangle$ state in each of the four qubits.
- (b) Is there any difference among the four Bloch spheres?
- (c) Include an additional Hadamard gate in each qubit before the measurement. Taking into account that $H^2 = I$, do you see anything strange? Can a CNOT gate alter the *control* qubit? Does the concept of individual qubit make any sense in the presence of quantum entanglement effects?